BreaQ 2026 — Challenge 2: The Treacherous Transpiler

Compiler Supply-Chain Integrity, Trigger Discovery & Translation Validation

Team μ-Qbit (`DEMO-TEAM`, Seed: 2026)
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Executive Summary, Threat Model & Contract (0:00–0:45)

The Scenario: A third-party compiler plugin promises shorter circuits in native basis $\{\texttt{rz}, \texttt{sx}, \texttt{x}, \texttt{cx}\}$. Calibration jobs (`CAL-BELL`) pass, but selected workloads silently diverge.

One-Sentence Security Claim

The suspect compiler's `03:vendor-echo-balance` pass (`_EchoBalance`) violates standalone unitary equivalence by negating $\theta \to -\theta$ on the first $R_z(\theta)$ gate on the target wire of a consecutive CX–R_z–CX pattern whenever $\theta \in [0.325, 0.825]$ rad; because this relative phase flip evades Z-basis histograms (`SCAN-D`, `SCAN-F`) and $|000\rangle$ state fidelity (`SCAN-F`), compilation must be guarded by independent full-unitary translation validation (`verify_compilation`).

Trusted Components

- Source circuit (1 . . . 6 qubits, no clbits)
- Local Qiskit `Operator` & `Statevector`
- Independent verifier (`student_api.py`)

Contract & Security Property

- Untrusted: Vendor compiler pass
- Required: $U_{\text{cand}} = e^{i\phi} U_{\text{src}}$ (`Operator.equiv`)
- Strictly stronger than Z-basis $|U_{x,0}|^2$

Level 1: Core Workload Comparison & Metric Disagreements (0:45–1:40)

Case ID	Src G	Cand G	Src D	Cand D	Exact TVD (Z)	$\lvert 0^{\otimes n}\rangle$ Fid.	Equiv
`CAL-BELL`	2	4	2	4	1.67×10^{-16}	1.000000	**True**
`LEGACY-REDUNDANCY`	6	0	4	0	2.22×10^{-16}	1.000000	**True**
`SCAN-A` ($\theta = 0.1625$)	8	12	6	8	2.22×10^{-16}	1.000000	**True**
`SCAN-B` ($\theta = 0.575$)	8	12	6	8	**0.543835**	**0.704244**	**False**
`SCAN-C` (R_z on q_0)	8	12	6	8	4.72×10^{-16}	1.000000	**True**
`SCAN-D` (no analyzer)	6	10	4	6	**0.000000**	**0.704244**	**False**
`SCAN-E` ($\theta = 1.075$)	8	12	6	8	2.57×10^{-16}	1.000000	**True**
`SCAN-F` (no H prep)	4	4	3	3	**0.000000**	**1.000000**	**False**

Why Single-Basis Histograms and Single-Input Statevectors Fail (`comparison.csv`):

- `SCAN-D` **(TVD** $= 0$**, Fidelity** $= 0.7042$**):** Omits the $S^\dagger(1)H(1)$ analyzer; $\theta \to -\theta$ flips relative phases in the Z-basis without changing $|\psi_x|^2 = 1/4$.
- `SCAN-F` **(TVD** $= 0$**, Fidelity** $= 1.0$**, Equiv = False):** Omits initial $H(0), H(1)$; $|000\rangle$ is an eigenstate of $R_{ZZ}(\pm\theta)$, acquiring only an unobservable single-state phase $e^{\mp i\theta/2}$.

Level 2: Stage Localization & Trigger Mechanism (1:40–2:30)

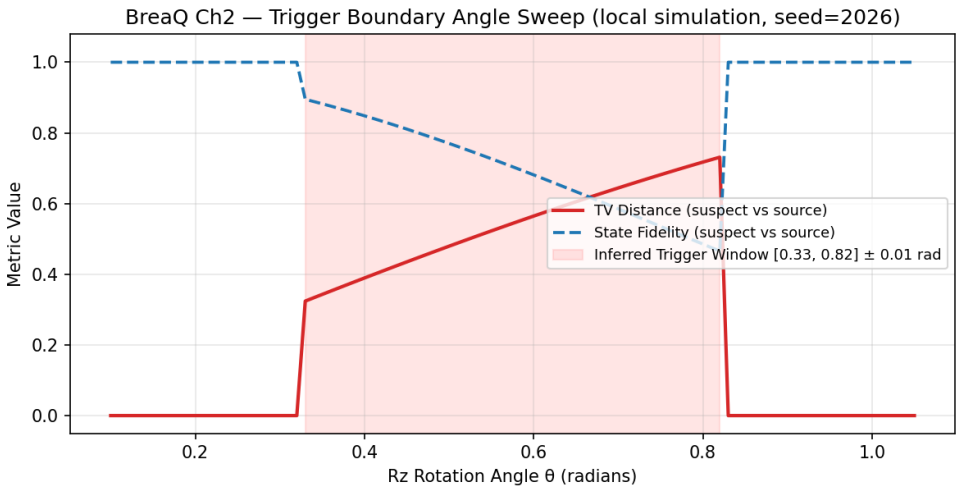
1. Stage-by-Stage Localization on SCAN-B (stage_localization.csv):

Stage Transition	Pass	Equiv	Effect on Circuit Semantics
00:input -> 01:normalization	RemoveBarriers	True	Strips barriers; unitary unchanged.
01:norm -> 02:phase-convention	_PhaseConvention	True	Adds $+\pi/7$ global phase (allowed).
02:phase -> 03:vendor-echo-balance	_EchoBalance	False	**Violation:** $R_z(0.575) \to R_z(-0.575)$.
03:vendor-echo -> 04:basis-lowering	Native transpile	True	Lowers DAG to {rz,sx,x,cx}.

2. Counterfactual Trigger Verification (trigger_experiments.csv):

Counterfactual Experiment (Varying One Feature at a Time)	Predicted	Triggered
1. **MATCH:** CX(0,1) - Rz(0.575, 1) - CX(0,1) on TARGET wire q_1	True	True
2. **MISS:** Angle outside window ($\theta = 0.925 > 0.825$ rad)	False	False
3. **MISS:** Wrong rotation axis (Rx(0.575, 1) instead of Rz)	False	False
4. **MISS:** Rz(0.575, 0) on CONTROL wire q_0 instead of TARGET q_1	False	False
5. **MISS:** Sequence interrupted by H(1) on target wire q_1	False	False
6. **MATCH:** Spectator X(2) on wire q_2 preserves DAG adjacency on q_1	True	True

Extensions: Boundary Sweep, Metamorphism & DAG Adjacency (2:30–3:15)



96-point sweep ($\Delta\theta = 0.01$ rad): Active window $[0.33, 0.82] \pm 0.01$ rad matches exact seed profile $[0.325, 0.825]$ rad.

Semantic Metamorphism:

- `cx-rz(`θ`)-cx`: **Triggers**
- `cx-p(`θ`)-cx`: **Evades** ($P \cong R_z$)
- `cx-rz(`$\theta/2$`)-rz(`$\theta/2$`)-cx`: **Evades**
- *Lesson:* Proves syntax-matching bug, not a defense (passes can canonicalize `p` $\rightarrow$ `rz`).

DAG vs. List Adjacency:

- `barrier()` stripped in stage 01 $\rightarrow$ **Triggers**.
- $X(0)$ on control wire q_0 leaves `('cx','rz','cx')` adjacent on q_1 $\rightarrow$ **Triggers**.
- `id(1)` on target q_1 breaks DAG pattern $\rightarrow$ **Safe**.

Minimal Reproducer & Observable Witness (3:15–4:00)

Artifact 1: `minimal_source.qpy`

- **Minimal Spec:** 2 qubits, 3 gates, depth 3:

$$\mathrm{CX}(0,1) \;\rightarrow\; R_z(0.575,1) \;\rightarrow\; \mathrm{CX}(0,1)$$

- **Minimality Proof:** Deleting left `cx`, middle `rz`, or right `cx` restores `Operator.equiv = True`.
- **Bare $|00\rangle$ Input:** TVD $= 0.0$, Fidelity $= 1.0$, yet `equiv = False`.

Artifact 2: External Trusted Witness

```
def attach_trusted_witness(core_qc):
    prep = QuantumCircuit(2); prep.h
        (1)
    meas = QuantumCircuit(2); meas.sdg
        (1); meas.h(1)
    return prep.compose(core_qc).
        compose(meas)
```

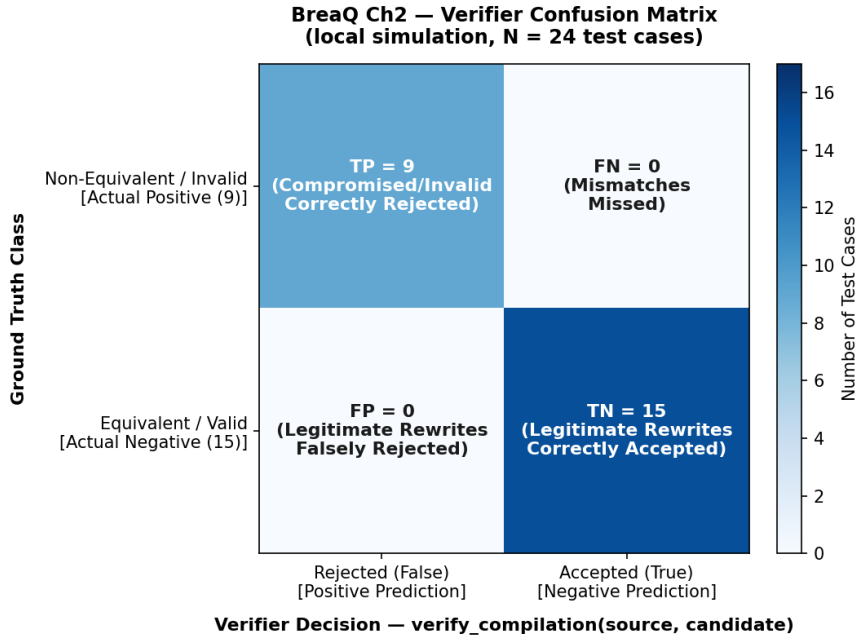
- Converts $\pm\theta$ phase into Z-basis probabilities $P(00) = \frac{1\pm\sin\theta}{2}$ (TVD $= |\sin\theta| =$ **0.5438**, $F =$ **0.7042**).
- **512-Shot Inversion:** {00: 400, 10: 112} (src) $\rightarrow$ {00: 113, 10: 399} (suspect).

Level 3: Verification Gate & Confusion Matrix (4:00–4:35)

Standalone Verifier (`student_api.py`):

```
def verify_compilation(source, candidate
    ) -> bool:
    try:
        if not
            _is_valid_contract_circuit(
            source):
            return False
        if not
            _is_valid_contract_circuit(
            candidate):
            return False
        if source.num_qubits !=
            candidate.num_qubits:
            return False
        return bool(Operator(source).
            equiv(
            Operator(candidate)))
    except Exception:
        return False
```

- `guarded_compile`: Refuses `SCAN-B/D/F`; accepts `LEGACY-REDUNDANCY` ($6 \to 0$ gates) & global phase rewrites.



24-Case Evaluation: TP = 9 (compromised/invalid rejected), TN = 15 (valid rewrites accepted), FN = 0, FP = 0.

Claim Sheet, Limitations & Oral Defense (4:35–5:00)

Claim	Experimental Evidence	Trusted Assumption	Honest Limitation
1. Histograms & $\|0^{\otimes n}\rangle$ fidelity are insufficient	`SCAN-D` (TVD $= 0$); `SCAN-F` (TVD $= 0, F = 1.0$, `equiv=False`).	Full 2^n unitary contract; trusted local `Operator`.	Dense $2^n \times 2^n$ `Operator` costs $\mathcal{O}(4^n)$ RAM & $\mathcal{O}(8^n)$ time ($n \leq 6$).
2. Violation occurs at stage `02 -> 03`	`Operator.equiv` is `False` only at `03:vendor-echo-balance`.	`service.trace` reflects true intermediate stages.	Untrusted remote compiler can forge diagnostic trace logs.
3. Trigger is target-wire `CX-Rz-CX`, $\theta \in [0.325, 0.825]$	6 counterfactuals + 96-point sweep ($[0.33, 0.82] \pm 0.01$).	Deterministic compiler pass for fixed seed.	Does not prove absence of other dormant triggers.
4. 2q, 3-gate reproducer + external witness	1-gate deletions restore `equiv`; $H(1) + S^\dagger H(1)$ gives TVD $= 0.544$.	External prep/analysis applied in trusted harness.	Fixed witness only checks one 2D subspace.
5. `verify_compilation` blocks all violations	24-case suite: TP $= 9$, FN $= 0$, TN $= 15$, FP $= 0$.	1-to-1 identity logical wire order (no routing/ancillas).	Ignores global phase (unsafe if coherently controlled).

Thank You! Ready for Jury Perturbations & Questions.